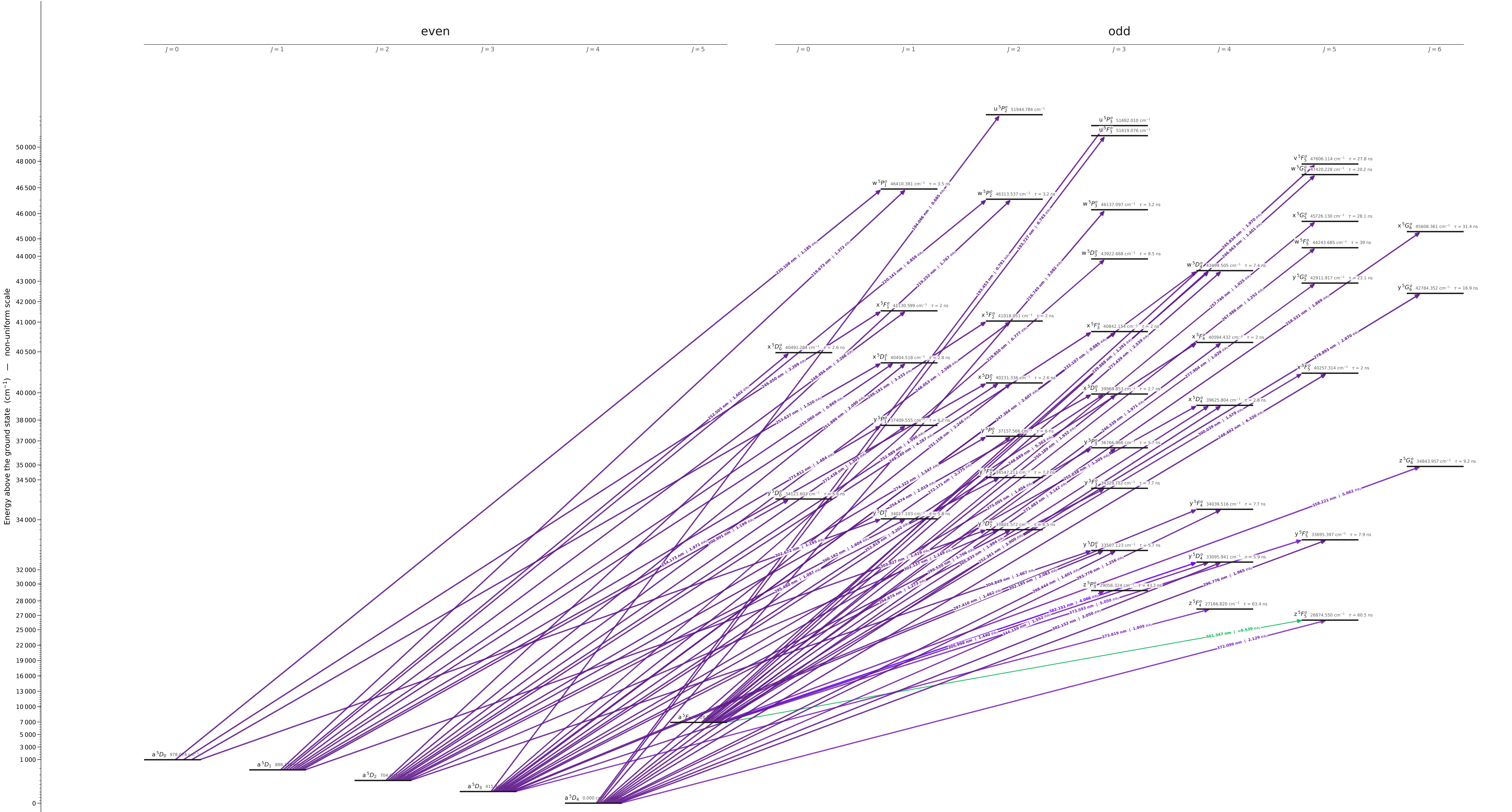
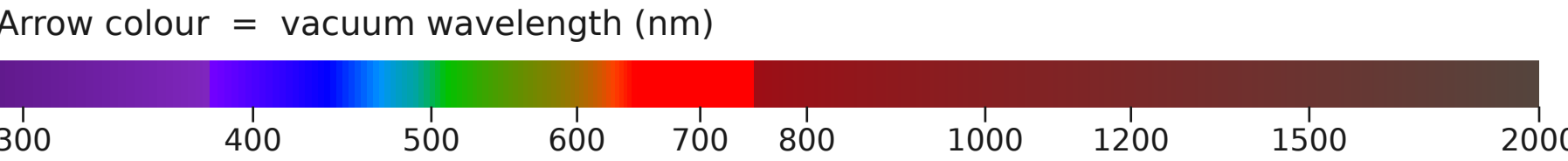


Iron

Fe (Fe I). Strongest classified transitions below 2 μm (NIST): 47 levels, 71 lines.

Fe⁺ ionisation limit 63737.70 cm⁻¹ = 7.90247 eV (156.893 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||e\mathbf{r}||J'\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e\mathbf{r} U'\rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁵ F ₅ – z ⁵ G ₆ ^o	358.2215	358.1193	836.8913	5.662	1.087e+08	17.3	100	358 nm laser-cooling line of iron
a ⁵ D ₄ – z ⁵ F ₅ ^o	372.0993	371.9934	805.6787	2.129	1.620e+07	2.631	99.6	372 nm resonance line: laser collimation / cooling of an...
a ⁵ F ₅ – z ⁵ F ₅ ^o	501.3466	501.2068	597.9745	≈0.539	4.251e+05	2.631	2.57	decay channel that completes optical pumping from the gro...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁵⁷Fe hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ⁵ D ₄	38.0795(10)	–	F=7/2: -95.20 F=9/2: +76.16
a ⁵ F ₅	87.246(3)	–	F=9/2: -261.74 F=11/2: +218.12
z ⁵ F ₅ ^o	81.7(9)	–	F=9/2: -245.07 F=11/2: +204.22
z ⁵ G ₆ ^o	31.24(5)	–	F=11/2: -109.34 F=13/2: +93.72

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a ⁵ F ₅ – z ⁵ G ₆ ^o	358.221	56-54	592.3(6)
a ⁵ F ₅ – z ⁵ G ₆ ^o	358.221	57-56	299.7(6)
a ⁵ F ₅ – z ⁵ G ₆ ^o	358.221	58-56	566.0(5)
a ⁵ D ₄ – z ⁵ F ₅ ^o	372.099	56-54	726(8)
a ⁵ D ₄ – z ⁵ F ₅ ^o	372.099	57-56	365(6)
a ⁵ D ₄ – z ⁵ F ₅ ^o	372.099	58-56	690(7)

Shift = ν(first isotope) – ν(second isotope).